



## SIMULATED REAL-TIME SENTIMENT ANALYSIS OF CUSTOMER FEEDBACK

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## ABSTRACT

This research presents the development of a sentiment analysis system with an interactive visual interface to improve the understanding of customer feedback. This work utilises the Recipe Reviews and User Feedback dataset, which is publicly available via the UCI Machine Learning Repository. The system applies a systematic NLP pipeline comprising data cleansing, token segmentation, and linguistic feature extraction for analysis. Since the dataset consists of unstructured textual reviews written in natural human language, NLP techniques are essential to convert this data into a format suitable for machine learning analysis. To capture the contextual meaning of customer reviews and the nuanced meanings in customer feedback, a pre-trained transformer model (BERT) is used to derive semantic-rich feature vectors from the text. Rather than using BERT as a standalone classifier, the model's contextualised word embeddings are used as numerical representations of the review text. The resulting embeddings, supplemented by basic text-based features, serve as input vectors for traditional supervised learning algorithms, specifically Logistic Regression and Linear SVC, to categorise reviews into positive, neutral, and negative sentiment classes. This hybrid approach enables the system to leverage the semantic understanding of transformer models while maintaining the interpretability and efficiency of traditional classifiers. An analysis of the dataset uncovered a significant positivity skew, a trend frequently seen in review-based data due to user self-selection and satisfaction bias. Both machine learning models demonstrated strong predictive performance, indicating that classical classifiers can perform effectively when enriched with contextual embeddings. Performance metrics such as accuracy, precision, recall, F1-score, and confusion matrices were utilized to assess model efficacy across sentiment categories rigorously. In addition to the predictive models, a simulated real-time sentiment monitoring component was developed. This interface allows users to manually input new reviews, which are processed immediately by the trained model and reflected in dynamic visual outputs, including sentiment count charts, trend lines, and a live prediction log. Although the real-time functionality operates in a simulated environment rather than through live API integration, it effectively demonstrates how sentiment analysis systems can be adapted for continuous monitoring scenarios.

This project demonstrates the potential of integrating Natural Language Processing, contextual vector representations, and classical classification methods with interactive visualisation tools to inform sentiment-based decision processes. Although the current version uses secondary data and manual inputs, the underlying design is adaptable to real-time environments through future integration of APIs, live data feeds, or fully end-to-end transformer-based models. The study thus reinforces the applied value of merging linguistic computation and machine learning within data-centric digital business contexts.

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## Chapter 1 Introduction

### 1.1 Introduction

The rapid growth of online platforms has significantly changed how businesses interact with customers, resulting in a continuous flow of feedback in the form of reviews, ratings, and comments. This large volume of textual feedback contains valuable information about customer satisfaction, preferences, and concerns. However, because this data is unstructured, analysing it manually is time-consuming and impractical. As a result, organisations increasingly rely on automated techniques to analyse customer feedback and support data-driven decision-making.

Sentiment analysis, an important subset of Natural Language Processing (NLP), focuses on automatically detecting and interpreting subjective expressions, such as opinions and emotional tone, in textual data. By converting human language into structured representations that machines can process, sentiment analysis allows organisations to monitor customer perceptions more efficiently. When combined with visual analytics, sentiment analysis can further support faster interpretation of trends and patterns in customer feedback. This study presents the development of a simulated real-time sentiment analysis and visualisation system using machine learning techniques. This study utilises the publicly available Recipe Reviews and User Feedback dataset from the UCI Machine Learning Repository as a secondary data source. Textual reviews are first processed using NLP techniques, after which a pretrained transformer-based language model (BERT) is used to generate contextual embeddings that capture the semantic meaning of customer reviews. The contextual embeddings serve as feature vectors for training supervised models, including Logistic Regression and Linear SVC, for classifying sentiment. In addition to offline model evaluation, the study implements a simulated real-time interface that allows new reviews to be manually entered and analysed instantly. The dashboard dynamically updates sentiment counts, trends, and interactive visual summaries, showcasing the system's potential for deployment in practical settings through integration with real-time data pipelines. This study demonstrates how integrating NLP methods, contextual embeddings, classical machine learning, and visual analytics can transform unstructured customer feedback into actionable insights for strategic business decision-making.

## **1.2 Background and Context**

Digital technologies have redefined how companies and customers interact, elevating feedback into a key strategic asset. Organisations now rely heavily on digital analytics to understand customer opinions and market perception. By applying natural language processing and machine learning techniques, businesses can interpret the emotional tone of customer reviews to assess satisfaction levels and identify potential improvements. Recent developments in AI, including real-time data processing and dynamic dashboard capabilities, have streamlined the handling of extensive customer feedback. Visual analytics tools help decision-makers quickly understand sentiment patterns and respond to emerging issues. In this context, sentiment analysis systems that combine predictive models with interactive visualisations play an important role in supporting customer-focused and data-driven business strategies.

## **1.2 Problem Statement**

Although sentiment analysis is a well-established research area, much of the existing work relies on static datasets and offline evaluation. This limits the ability to observe how customer sentiment changes over time. Many sentiment analysis dashboards function as post-analysis tools rather than as systems that support near-real-time decision-making. As a result, emerging trends or early signs of dissatisfaction may not be identified quickly enough. There is also a gap between advanced NLP-based sentiment modelling and interactive visual analytics that update continuously. This gap makes it difficult for organisations to convert large volumes of unstructured text into timely and actionable insights. This project addresses part of this challenge by developing an integrated sentiment analysis and visualisation system that supports simulated real-time monitoring. New reviews can be manually entered and analysed instantly, with the dashboard updating sentiment distributions and trends in real time. While live data streaming is not implemented, the system provides a foundation for future real-time applications. This study, therefore, explores modern NLP techniques that address the limitations of traditional feature representations by incorporating contextual language understanding.

## **1.4 Research Aim and Objectives**



**Aim:** This study aims to develop a simulated real-time sentiment analysis and visualisation system to analyse product feedback using the *Recipe Reviews and User Feedback* dataset from the UCI Machine Learning Repository.

### **Objectives**

- To preprocess and analyse textual feedback using Natural Language Processing (NLP) and sentiment classification models
- To develop a simulated real-time dashboard to visualise sentiment trends and customer insights
- To evaluate the accuracy, responsiveness, and reliability of the system through experimental testing
- To identify practical business implications for managing customer experience

## **1.5 Research Rationale**

### ***What is the issue?***

In the current digital environment, organisations receive large volumes of unstructured customer feedback through reviews and online interactions. However, most sentiment analysis systems still operate on static datasets, meaning insights are generated only after data collection. This delay prevents organisations from responding quickly to changing customer opinions.

### ***Why is this an issue?***

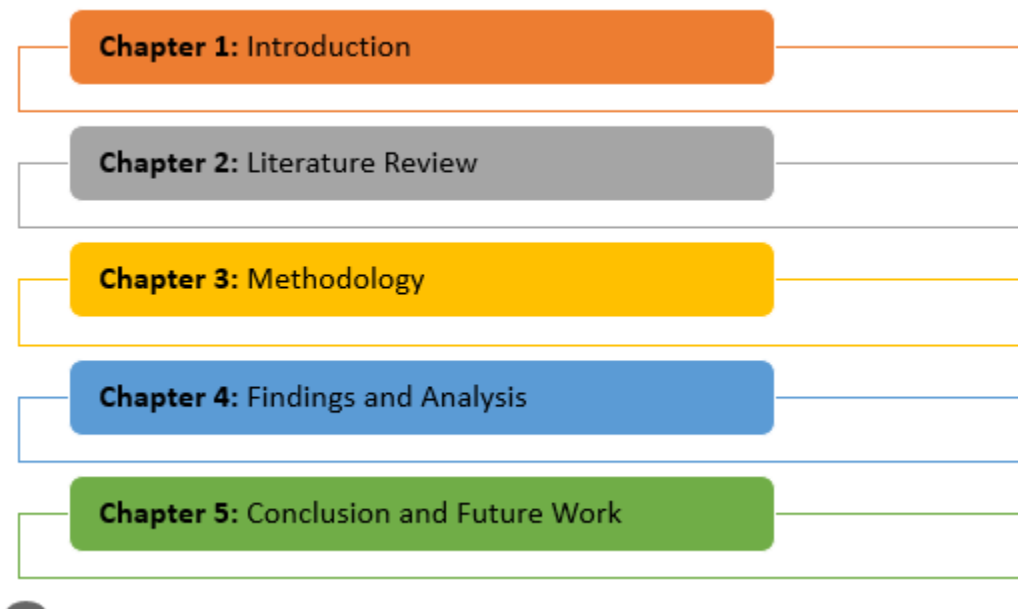
Lack of real-time understanding means a business will not react quickly to customers' shifting emotions, which could lead to missed opportunities, customer dissatisfaction, and poor market positioning. This lack of integration prevents businesses from turning customer feedback into timely and meaningful decisions.

### ***How can this issue be solved?***

This study proposes a solution that combines NLP, machine learning, and visual analytics in a simulated real-time environment. The developed sentiment dashboard transforms unstructured feedback into visual insights, enabling users to track sentiment patterns, identify emerging issues,

and support informed decision-making. Although real-time data streaming is simulated, the system demonstrates how such an approach could be extended to real-world applications.

## 1.6 Outline of the dissertation



**Figure 1.1: Research Framework**

(Source: Created in MS Word)

The dissertation is divided into five major chapters to provide a detailed overview of the research process and results.

**Chapter 1:** Introduction presents the background, problem statement, research aims, and significance of the study.

**Chapter 2:** Literature Review reviews existing research related to sentiment analysis, NLP techniques, and visual analytics.

**Chapter 3:** Methodology describes the dataset selection, preprocessing steps, sentiment modelling approach, and dashboard development.

**Chapter 4:** Findings and Analysis presents the experimental results, model performance evaluation, and dashboard analysis.

**Chapter 5:** Conclusion and Future Work summarises the findings, discusses limitations, and outlines directions for future research

This chapter sets the context, purpose and meaning of the research. The following chapter reviews the theoretical and empirical background of sentiment analysis, natural language processing, and dashboard visualisation.

## **1.7 Summary**

This chapter introduced the research on simulated real-time sentiment analysis and dashboard visualization for customer feedback. It outlined the background, problem statement, research aims and objectives, and the rationale behind the study. The importance of integrating sentiment analysis with visual analytics to enable faster, more informed decision-making was highlighted. Finally, the structure of the dissertation was presented, offering a roadmap for the chapters that follow.

## Chapter 2: Literature Review

### 2.1 Introduction

The rapid growth of user-generated textual data across digital platforms has increased demand for automated techniques to extract meaningful insights from unstructured text. Sentiment analysis, a key application of Natural Language Processing (NLP), focuses on identifying opinions, emotions, and attitudes expressed in written language. It has become a widely adopted approach for analysing online reviews and feedback data.

This chapter critically reviews the existing literature on NLP preprocessing, transformer-based language models, and classical supervised machine learning approaches for sentiment classification. Particular emphasis is placed on hybrid sentiment analysis frameworks that combine contextual embeddings with traditional classifiers, as this approach aligns directly with the analytical pipeline implemented in the present study. By synthesising prior research, this chapter establishes a theoretical foundation for the methodological choices adopted in this applied computing research project.

### 2.2 Natural Language Processing and Text Preprocessing

Natural Language Processing enables computational systems to analyse and interpret human language by converting raw textual input into structured representations suitable for machine learning. *Jurafsky and Martin (2023)* described real-world textual data as inherently noisy and inconsistent, often containing variations in grammar, spelling, punctuation, and formatting. As a result, preprocessing is a fundamental requirement before textual data can be effectively modelled. They emphasised that preprocessing techniques such as tokenisation, text normalisation, stop-word removal, and punctuation handling reduce linguistic variability and improve the reliability of textual representations. In sentiment analysis tasks, these steps help ensure that irrelevant components do not dominate the learning process. *Pang and Lee (2008)* highlighted that preprocessing plays a critical role in sentiment classification, particularly for review-based datasets, where informal language and repetitive expressions are standard. They noted that consistent feature representation improves classification performance by reducing unnecessary dimensionality.

Although transformer-based models can capture contextual meaning, Devlin et al. (2019) noted that they still require well-formed textual input. Clean and standardised text supports stable tokenisation and contributes to reliable embedding generation. Consequently, modern NLP pipelines typically incorporate basic preprocessing prior to feature extraction, a practice followed in the present study. TF-IDF has been widely applied in sentiment analysis due to its simplicity and effectiveness; however, it fails to capture contextual and semantic relationships between words (*Salton & Buckley, 1988; Sebastiani, 2002*).

## 2.3 Classical Machine Learning Models for Sentiment Classification

Supervised machine learning algorithms have been widely applied to sentiment classification due to their ability to learn decision boundaries from labelled data. Logistic Regression remains a widely used classifier in text analytics due to its simplicity, efficiency, and interpretability. *Ng and Jordan (2002)* compared discriminative and generative learning approaches and demonstrated that discriminative models, such as Logistic Regression, perform effectively when informative feature representations are available. They argued that Logistic Regression directly models the relationship between input features and class labels, making it suitable for classification tasks involving high-dimensional textual data. As a result, Logistic Regression continues to serve as a strong baseline model in sentiment analysis research. Support Vector Machines have also been extensively used for text classification. *Joachims (1998)* demonstrated that SVMs are particularly effective for text categorisation due to their margin-maximisation principle, which enables strong generalisation in sparse, high-dimensional feature spaces. He showed that SVMs perform well even when the number of features exceeds the number of training samples, a common characteristic of textual data. Despite their effectiveness, both Logistic Regression and SVMs rely heavily on the quality of feature representations. When combined with simple word-based features, these models may fail to capture deeper semantic relationships within text. This limitation has motivated research into richer feature extraction techniques, particularly contextual embeddings.

## 2.4 Transformer-Based Language Models and Contextual Embeddings

Transformer-based architectures represent a significant advancement in NLP by enabling models to capture contextual relationships within text. *Devlin et al. (2019)* introduced Bidirectional Encoder Representations from Transformers (BERT), a language model that learns bidirectional representations by conditioning on both preceding and following words simultaneously. This allows words to be represented differently depending on context, which is particularly important for sentiment analysis. *Devlin et al.* reported that contextual embeddings generated by BERT significantly improve performance across a range of NLP tasks, including text classification. However, they also noted that fine-tuning large transformer models can be computationally expensive and resource-intensive, which may not always be practical in applied research environments. To address this, *Sun et al. (2019)* examined alternative strategies for applying BERT to classification tasks. They showed that pretrained BERT representations can be effectively used as feature inputs for downstream classifiers. This approach enables models to leverage contextual semantics while maintaining simpler, more efficient training pipelines.

## 2.5 Hybrid Sentiment Analysis Approaches

Hybrid sentiment analysis approaches combine transformer-based contextual embeddings with traditional supervised machine learning classifiers. This strategy has gained attention because it balances semantic richness with computational efficiency and interpretability. *Reimers and Gurevych (2019)* demonstrated that sentence-level embeddings derived from transformer models preserve semantic information and can be effectively utilised by conventional classifiers.

Hybrid pipelines are particularly suitable for applied computing research, where transparency and resource constraints are important considerations. Unlike end-to-end deep learning systems, hybrid approaches allow researchers to analyse classifier behaviour more clearly while still benefiting from deep contextual representations. *Sun et al. (2019)* further highlighted that hybrid architectures are practical for real-world text classification tasks, particularly when complete model fine-tuning is unnecessary. The present study adopts this approach by generating contextual embeddings using a pretrained BERT model and applying Logistic Regression and Support Vector Machine classifiers for sentiment prediction.

## 2.6 Model Evaluation in Sentiment Analysis

Reliable evaluation of sentiment classification models requires the use of multiple performance metrics. Sokolova and Lapalme (2009) argued that accuracy alone can be misleading, particularly in multi-class or imbalanced classification problems. They recommended evaluating models using precision, recall, F1-score, and confusion matrices to obtain a comprehensive understanding of classification performance. Precision and recall provide insight into different types of classification errors, while confusion matrices allow detailed examination of misclassification patterns across sentiment classes. This multi-metric evaluation strategy has become standard practice in sentiment analysis research and was adopted in the present study to ensure transparent and reliable assessment of model performance.

## **2.7 Dimensionality Reduction Using Principal Component Analysis (PCA)**

High-dimensional feature representations are common in text-based sentiment analysis, particularly when contextual embeddings such as those generated by transformer models are used. While these embeddings capture rich semantic information, they can increase computational complexity and introduce redundancy within the feature space. Principal Component Analysis (PCA) is a widely used dimensionality reduction technique that transforms high-dimensional data into a lower-dimensional representation while preserving as much variance as possible (*Jolliffe, 2002*). By projecting data onto orthogonal components, PCA reduces noise and correlation among features, leading to more stable and efficient model training. *Abdi and Williams (2010)* explained that PCA is especially effective when applied before supervised learning, as it can improve computational efficiency without significantly affecting predictive performance. In sentiment analysis tasks, PCA has been used to reduce embedding dimensionality and enhance classifier robustness, particularly when working with large-scale text representations. Given the high-dimensional nature of BERT-generated embeddings, PCA provides a practical solution for managing feature complexity while retaining meaningful information. This approach aligns with applied machine learning practices and supports efficient sentiment classification.

## **2.8 Model Generalisation and Overfitting in Machine Learning**

Model generalisation refers to the model's ability to perform well on unseen data. Overfitting occurs when a model learns patterns specific to the training data rather than capturing generalisable relationships, resulting in reduced performance on new inputs.

*Hastie, T., Tibshirani, R., Friedman, J. (2009)* emphasised that overfitting is a common challenge in supervised learning, especially when models are trained on high-dimensional feature spaces. They recommended evaluating both training and testing performance to assess whether a model generalises effectively. Comparing training accuracy with test accuracy is a widely adopted method for detecting overfitting. A small gap between these values indicates that the model has learned meaningful patterns rather than memorising the training data (*Goodfellow et al., 2016*). In sentiment classification tasks, this approach is particularly significant given class imbalance and linguistic variability within textual data. Therefore, overfitting analysis is a critical component of reliable sentiment modelling and supports the use of generalisation-focused evaluation strategies.

## 2.9 Literature Gap

Although sentiment analysis has been widely studied, gaps remain in applied contexts. Many studies focus either on fully deep learning-based architectures or on traditional machine learning approaches using simple text representations. Compared with other studies, comparatively few examine hybrid pipelines that integrate transformer-based embeddings with classical classifiers in resource-constrained environments where interpretability and efficiency are critical. Furthermore, sentiment analysis research frequently emphasises social media datasets, while review-based datasets remain highly relevant for business analytics and decision-making but receive comparatively less attention. Addressing these gaps through practical hybrid sentiment analysis frameworks contributes to the applied relevance of NLP research in computing science.

## 2.10 Summary

This chapter reviewed literature related to NLP preprocessing, classical machine learning classifiers, transformer-based language models, hybrid sentiment analysis approaches, and evaluation strategies. Prior research supports the use of contextual embeddings combined with



traditional classifiers as a practical and effective solution for sentiment classification. These findings provide a strong theoretical foundation for the analytical design implemented in the present study.

## **Chapter 3 Methodology**

### **3.1 Chapter Overview**

This chapter explains the methodology used to build, implement, and assess a simulated real-time sentiment analysis and visualisation system. The approach taken in this study supports the main research objective of analysing customer review text and generating sentiment insights using Natural Language Processing (NLP), transformer-based text representations, and supervised machine learning techniques. The process begins with a secondary dataset containing customer reviews. The text data is prepared through basic cleaning and normalisation to make it suitable for analysis. Context-aware numerical representations are then generated using a pretrained BERT embedding model, transforming textual reviews into feature vectors that machine learning algorithms can process. These embeddings are used to train supervised classifiers, specifically Logistic Regression and Linear Support Vector Machine (Linear SVC), to categorise reviews into positive, neutral, and negative sentiment classes. Model performance is assessed using standard classification metrics, with particular emphasis on the macro-averaged F1 score to ensure balanced evaluation across all sentiment classes. Finally, the selected model is integrated into a simulated real-time system that allows users to input new reviews and view instant sentiment predictions and updated visual outputs.

### **3.2 Research Philosophy**

This study follows an objective, data-driven approach, with conclusions based on measurable, observable results. Although customer reviews are written in natural language, the analysis converts them into numerical form for processing with machine learning techniques. Customer sentiment is identified using NLP and machine learning models rather than personal judgment. Model performance is evaluated using standard metrics such as accuracy, precision, recall, and F1-score. By clearly describing the data preprocessing steps, model settings, and evaluation methods, the study ensures that the results are consistent, reliable, and reproducible.

### **3.3 Research Approach**

A deductive research approach is adopted, where established theories and techniques from sentiment analysis and Natural Language Processing are applied to a specific problem scenario. Existing research suggests that customer opinions expressed in text contain identifiable linguistic patterns that correspond to sentiment polarity. Based on this assumption, known preprocessing techniques, transformer-based embedding methods, and supervised machine learning classifiers are applied to classify sentiment in customer reviews. The deductive approach allows the study to test and validate established techniques within a simulated real-time sentiment analysis context.

### **3.4 Research Design**

This study adopts a quantitative, applied research design to build and evaluate a practical sentiment analysis system for customer reviews. The research follows an artefact-driven approach, where the primary outcome is a working prototype that combines Natural Language Processing (NLP), machine learning techniques, and visual analytics. The system is developed using a clear and structured workflow. First, the customer review text is pre-processed to remove noise and prepare it for analysis. Next, contextual features are generated using pre-trained BERT embeddings, which transform the text into meaningful numerical representations. These embeddings are then used to train supervised machine learning models, specifically Logistic Regression and a Linear Support Vector Machine (Linear SVC), to classify reviews into sentiment categories. Model performance is evaluated using standard classification metrics, including accuracy, precision, recall, and F1 Score. However, greater importance is given to the macro-averaged F1-score to ensure that all sentiment classes are evaluated fairly, particularly in the presence of class imbalance. To demonstrate practical applicability, a simulated real-time component is included, allowing new reviews to be analysed instantly and visualised through an interactive dashboard. Overall, this research design supports hands-on experimentation while closely reflecting how sentiment monitoring systems are applied in real-world scenarios.

### 3.5 Dataset Collection Approach & Selection Rationale

| Recipe ID | User ID | Review Text                 | Rating | Sentiment |
|-----------|---------|-----------------------------|--------|-----------|
| 1         | 14299   | Creamy Vag... [Review text] | 5      | Positive  |
| 2         | 14299   | Creamy Vag... [Review text] | 5      | Positive  |
| 3         | 14299   | Creamy Vag... [Review text] | 5      | Positive  |
| 4         | 14299   | Creamy Vag... [Review text] | 5      | Positive  |
| 5         | 14299   | Creamy Vag... [Review text] | 5      | Positive  |
| 6         | 14299   | Creamy Vag... [Review text] | 5      | Positive  |
| 7         | 14299   | Creamy Vag... [Review text] | 5      | Positive  |
| 8         | 14299   | Creamy Vag... [Review text] | 5      | Positive  |
| 9         | 14299   | Creamy Vag... [Review text] | 5      | Positive  |
| 10        | 14299   | Creamy Vag... [Review text] | 5      | Positive  |
| 11        | 14299   | Creamy Vag... [Review text] | 5      | Positive  |
| 12        | 14299   | Creamy Vag... [Review text] | 5      | Positive  |
| 13        | 14299   | Creamy Vag... [Review text] | 5      | Positive  |
| 14        | 14299   | Creamy Vag... [Review text] | 5      | Positive  |
| 15        | 14299   | Creamy Vag... [Review text] | 5      | Positive  |
| 16        | 14299   | Creamy Vag... [Review text] | 5      | Positive  |
| 17        | 14299   | Creamy Vag... [Review text] | 5      | Positive  |
| 18        | 14299   | Creamy Vag... [Review text] | 5      | Positive  |
| 19        | 14299   | Creamy Vag... [Review text] | 5      | Positive  |
| 20        | 14299   | Creamy Vag... [Review text] | 5      | Positive  |
| 21        | 14299   | Creamy Vag... [Review text] | 5      | Positive  |
| 22        | 14299   | Creamy Vag... [Review text] | 5      | Positive  |
| 23        | 14299   | Creamy Vag... [Review text] | 5      | Positive  |
| 24        | 14299   | Creamy Vag... [Review text] | 5      | Positive  |
| 25        | 14299   | Creamy Vag... [Review text] | 5      | Positive  |
| 26        | 14299   | Creamy Vag... [Review text] | 5      | Positive  |
| 27        | 14299   | Creamy Vag... [Review text] | 5      | Positive  |
| 28        | 14299   | Creamy Vag... [Review text] | 5      | Positive  |
| 29        | 14299   | Creamy Vag... [Review text] | 5      | Positive  |
| 30        | 14299   | Creamy Vag... [Review text] | 5      | Positive  |
| 31        | 14299   | Creamy Vag... [Review text] | 5      | Positive  |
| 32        | 14299   | Creamy Vag... [Review text] | 5      | Positive  |
| 33        | 14299   | Creamy Vag... [Review text] | 5      | Positive  |
| 34        | 14299   | Creamy Vag... [Review text] | 5      | Positive  |
| 35        | 14299   | Creamy Vag... [Review text] | 5      | Positive  |
| 36        | 14299   | Creamy Vag... [Review text] | 5      | Positive  |
| 37        | 14299   | Creamy Vag... [Review text] | 5      | Positive  |
| 38        | 14299   | Creamy Vag... [Review text] | 5      | Positive  |

Figure 3.1 Dataset View

The dataset used in this study is the Recipe Reviews and User Feedback dataset sourced from the UCI Machine Learning Repository. The dataset contains approximately 18,000 user-generated reviews of food recipes, along with associated metadata such as ratings and feedback. These reviews represent genuine consumer opinions, making the dataset well-suited for sentiment analysis tasks. The dataset is treated as secondary data, with review text serving as the primary input for NLP and sentiment modelling. As the data is publicly available and anonymised, it is appropriate for academic research. The dataset's size and diversity provide sufficient variability to effectively train and evaluate supervised machine learning models.

### 3.6 Data Pre-processing

The dataset was prepared using basic NLP preprocessing to support data cleaning and exploratory analysis. Unnecessary columns were removed, and records containing missing or invalid review text were excluded to ensure data quality. The review text was standardised by converting it to lowercase and removing URLs, memorable characters, and numerical values. Tokenisation, stopword removal, and lemmatisation were applied solely for exploratory analysis and visualisation tasks, such as generating word clouds, to identify standard terms and themes within the dataset.

For sentiment modelling, the original review text, without lemmatisation or stopword removal, was used to generate contextual embeddings using a pretrained BERT model. BERT performs

internal subword tokenisation and is designed to operate on minimally processed text. The resulting embeddings were used as numerical feature inputs for supervised machine learning classifiers.

### 3.7 Sentiment Modelling Architecture

The sentiment analysis system follows a **hybrid approach**, combining transformer-based text embeddings with traditional machine learning models. This design was chosen to achieve a balance between accuracy, efficiency, and interpretability.

The system consists of the following steps:

- Cleaning and preparing review text
- Converting text into contextual embeddings using a pretrained BERT model
- Classifying sentiment using supervised machine learning models
- Displaying results through a simulated real-time dashboard

This approach ensures that meaningful information from the text is captured while enabling fast, understandable sentiment predictions.

### 3.8 BERT Embedding Generation

A RoBERTa-based transformer model from CardiffNLP is used as the embedding extractor via Hugging Face. Instead of training a deep learning model from scratch, the pre-trained BERT model was used as a feature extractor to convert each review into a numerical representation of its meaning. These embeddings capture contextual information from the text and were then used as input features for supervised machine learning models to perform sentiment classification.

### 3.9 Dimensionality Reduction Using PCA

After generating contextual embeddings with the pre-trained BERT model, Principal Component Analysis (PCA) was applied to reduce feature dimensionality. Since BERT embeddings are high-dimensional, PCA helps remove redundancy while preserving the most important information. This reduction improves computational efficiency and supports more stable training of the machine learning classifiers. The use of PCA follows standard practices for managing high-dimensional numerical features in applied machine learning tasks.

### 3.10 Supervised Machine Learning Models

Sentiment classification was performed using two supervised machine learning models applied to the BERT-generated embeddings.

**Logistic Regression** was used as a baseline classifier due to its efficiency and suitability for high-dimensional feature spaces.

**A Linear Support Vector Machine (Linear SVC)** was employed as a comparison model, as it is effective for text classification tasks involving high-dimensional data.

Both models were trained on the exact embedding representations to ensure a fair comparison. The final selected model was saved using the Joblib library to support reuse and deployment without retraining.

### 3.11 Overfitting Detection and Model Generalisation

Model generalisation was assessed by comparing training and testing accuracy for each classifier. The difference between these values, referred to as the overfitting gap, was used to identify potential overfitting. A small gap indicates good generalisation to unseen data, while a larger gap suggests overfitting. This approach follows standard machine learning practice and helps ensure that the models learn generalisable patterns rather than memorising the training data.

### 3.12 Model Evaluation

Model performance is evaluated using standard classification metrics, including accuracy, precision, recall, F1-score, and confusion matrices. While accuracy provides an overall measure of correct predictions, precision and recall assess class-wise performance. The F1-score, particularly the macro-averaged F1-score, is used as the primary metric to ensure balanced evaluation across all sentiment classes. Confusion matrices are employed to analyse misclassification patterns and identify areas where the models perform well or require improvement.

### 3.13 Simulated Real-Time Sentiment Analysis and Visualisation

The trained sentiment model is used in a simulated real-time setting where users can manually enter new review text. Each input is processed through the same preprocessing, embedding, and classification steps, and the predicted sentiment is displayed immediately with updated visual summaries. Although live data streams are not used, this setup demonstrates how real-time sentiment analysis could be implemented in future systems.

### **3.14 Dashboard Development Process**

The dashboard was developed to present sentiment analysis results in a transparent and interpretable visual format. After the classification models generated sentiment predictions, the outputs were visualised using Python-based libraries such as Matplotlib and Seaborn. The dashboard includes visual elements, such as sentiment distribution charts and trend plots, that illustrate overall patterns in customer feedback. In addition, a word cloud was generated from the review text to highlight the most frequently occurring words across the dataset. This word cloud provides a high-level overview of commonly discussed terms and supports qualitative interpretation of customer opinions. To demonstrate real-time analytical behaviour, a simulated real-time interface was implemented that allows users to enter new reviews and immediately observe the predicted sentiment manually and updated visual summaries. Although the dashboard operates within a notebook environment and does not use live data APIs, it effectively demonstrates how sentiment outputs can be monitored and visualised dynamically. Overall, the dashboard serves as a lightweight decision-support tool that enhances the interpretability of sentiment analysis results and illustrates how similar systems could be extended in the future using automated data streams and web-based visualisation frameworks.

### **3.15 Ethical Considerations**

This study uses a publicly available, anonymised dataset from the UCI Machine Learning Repository, with no personally identifiable information. Ethical standards are maintained through transparent documentation of data preprocessing, model selection, and evaluation procedures to ensure reproducibility and fairness. Multiple evaluation metrics are applied to minimise potential bias, and all analysis is conducted in accordance with academic integrity guidelines.

### **3.16 Chapter Summary**

This chapter outlined the methodology used to develop the sentiment analysis and dashboard system. It discussed the research approach, dataset selection, data preprocessing, sentiment modelling techniques, and the use of NLP and machine learning for sentiment classification. The chapter also described the visualisation of sentiment results through an interactive dashboard and addressed ethical considerations related to data usage and transparency. The next chapter presents the system implementation, experimental results, and performance analysis.



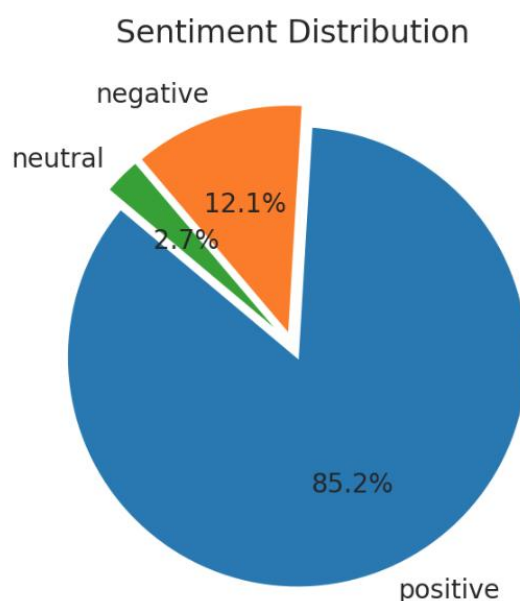
## Chapter 4: Findings and Analysis

### 4.1 Introduction

This chapter presents the findings and analysis of the sentiment analysis system developed in this study. It examines patterns identified through exploratory data analysis, evaluates the performance of supervised sentiment classification models, and assesses the effectiveness of the simulated real-time dashboard. The chapter concludes with a discussion of the key results and their implications

### 4.2 Exploratory Data Analysis and Visualisations

Exploratory Data Analysis (EDA) was conducted to understand the structure, distribution, and general characteristics of the recipe review dataset before applying sentiment classification models.

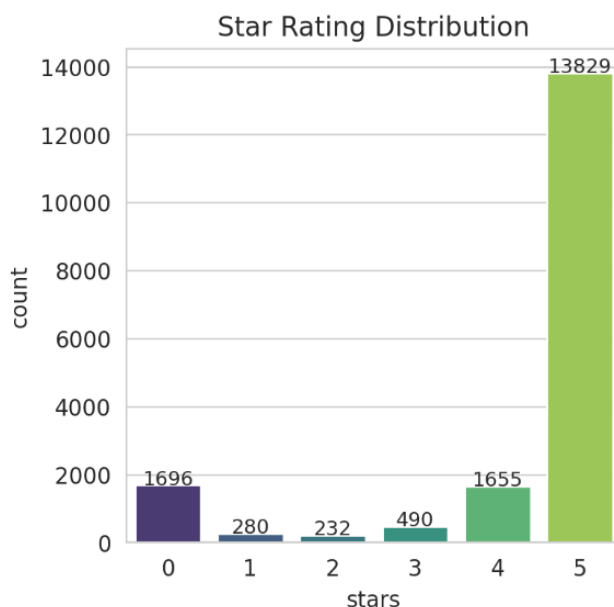


**Figure 4.1: Sentiment Distribution of Reviews**

(Source: Created in Google Colab)

Figure 4.1 illustrates the distribution of sentiment classes within the dataset. The results show a strong imbalance in sentiment, with positive reviews accounting for approximately 85.2% of the total data. Negative reviews account for around 12.1%, while neutral reviews account for only

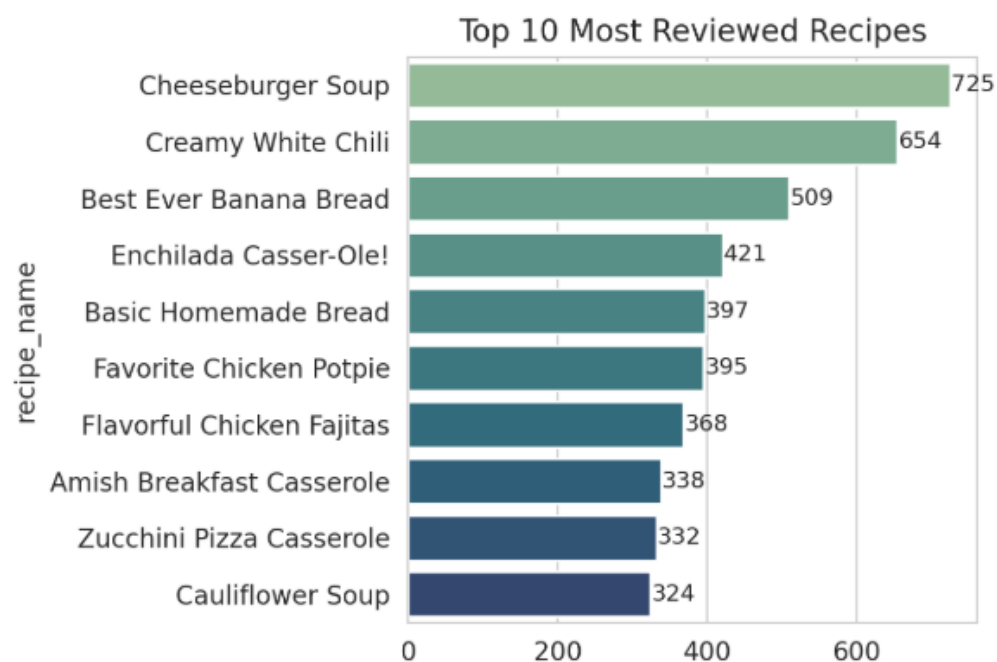
2.7% of the dataset. This imbalance is commonly observed in user-generated review platforms, where users are more likely to share positive experiences than neutral or negative ones. The presence of class imbalance highlights the limitation of relying solely on accuracy and supports the use of additional evaluation metrics such as precision, recall, and F1-score.



**Figure 4.2: Star Rating Frequency Chart**

(Source: Created in Google Colab)

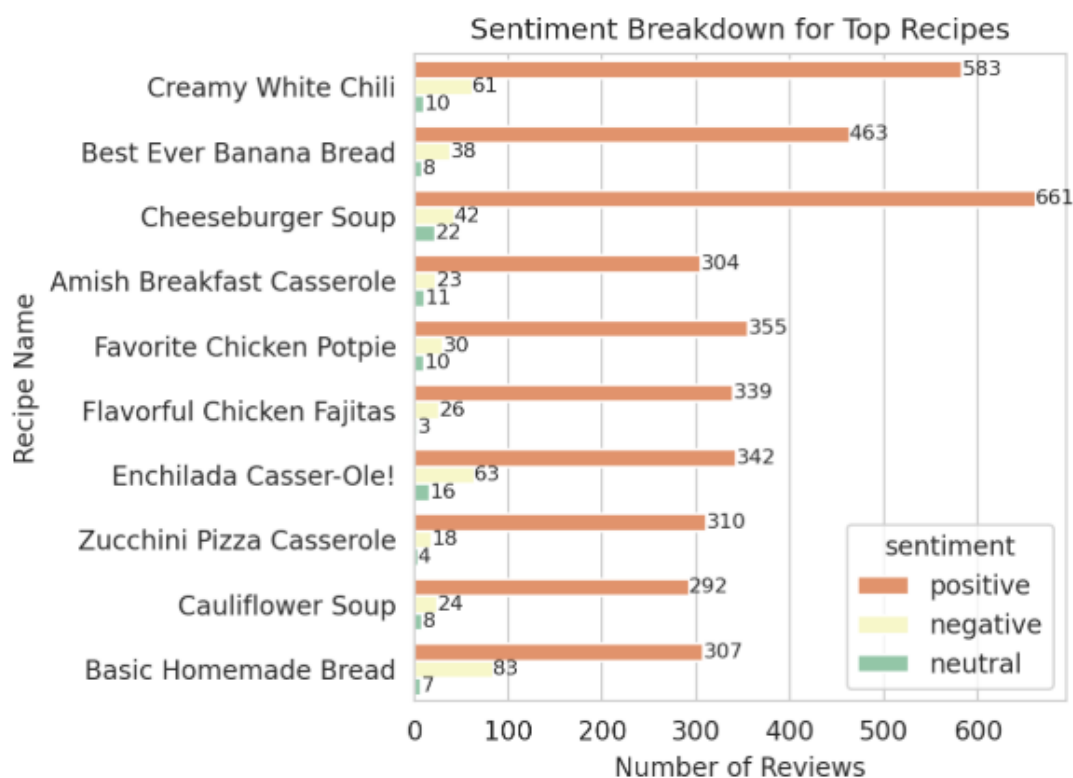
The distribution of star ratings reflects sentiment and is heavily skewed towards higher ratings. Figure 4.2 reveals that 5-star reviews are the most common, with 13,829 entries, followed by 4-star (1,655) and 3-star (490). It has lower ratings (1 and 2 stars) that occur rarely. Such similarity between the numerical rating and the labelled sentiment reinforces the validity of the sentiment mapping process. The spread also indicates that user satisfaction is consistently high on the platform, but the few low ratings complicate the identification of subtle trends related to negative sentiment.



**Figure 4.3: Top 10 Most Reviewed Recipes**

(Source: Created in Google Colab)

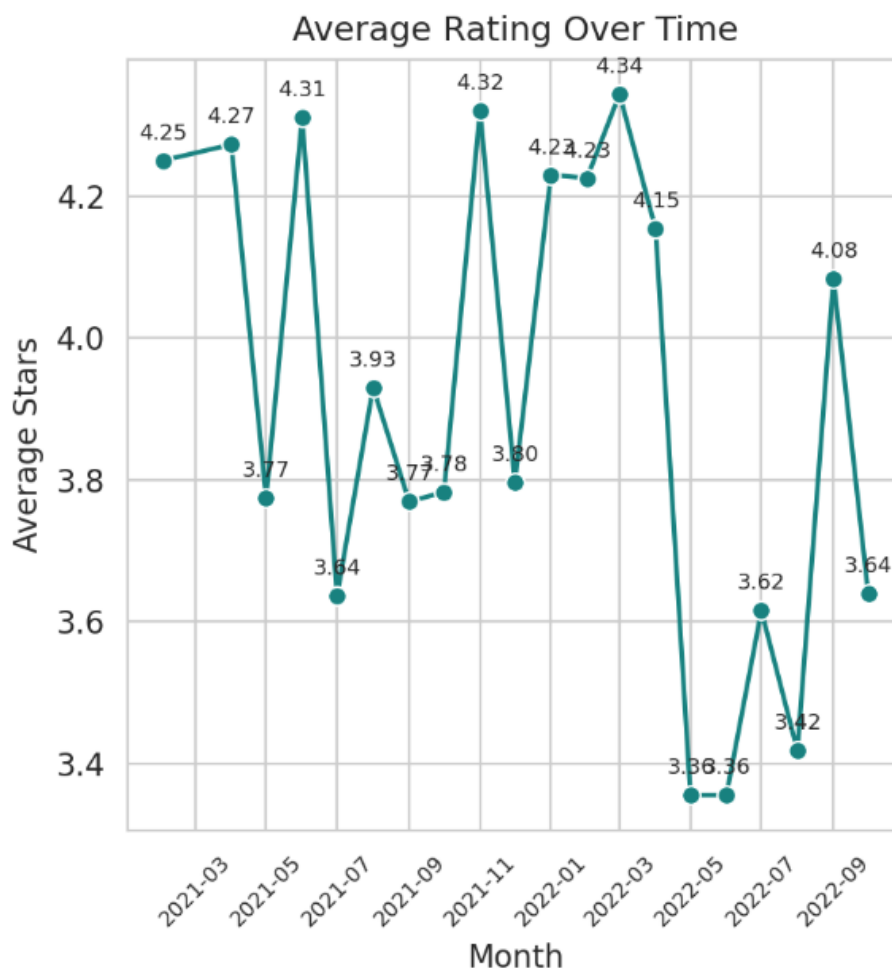
Figure 4.3 shows the most popular recipes, providing insight into how users are engaging. Meals like Cheeseburger Soup (725 reviews) and Creamy White Chilli (654 reviews) are the most interactive meals. The other well-reviewed dishes, such as Best Ever Banana Bread and Enchilada Casser-Ole, also show high interest in the community. These are typical dishes that tend to serve as references whenever a new or related dish is to be made. Large volumes of reviews also reflect increased use and a variety of opinion sources, which may increase the representation of sentiment in these recipes.



**Figure 4.4: Sentiment Breakdown by Recipe**

(Source: Created in Google Colab)

Figure 4.4 demonstrates that even the most-reviewed recipes have mostly positive sentiment. As an example, Cheeseburger Soup has 661 positive reviews, with few negative or neutral ones. Although there are recipes where the degree of neutral or negative feedback is somewhat greater, the overall tendency aligns with the dataset-wide sentiment imbalance. This discussion shows that popularity in recipes does not necessarily contribute to an increase in criticism, but popular foods are likely to continue to have high levels of satisfaction in different groups of users.



**Figure 4.5: Temporal Variation in Recipe Ratings**

(Source: Created in Google Colab)

Figure 4.5 illustrates changes in average star ratings over time. The average ratings generally fluctuate between 3.3 and 4.3 throughout the observed period. A noticeable decline appears during mid-2022, which may be influenced by seasonal factors, changes in user expectations, or variations in the types of recipes being reviewed. These findings indicate that sentiment can vary over time, highlighting the potential for future work involving temporal or seasonal sentiment analysis.



### Figure 4.6 Word Clouds for Sentiment Classes

(Source: Created in Google Colab)

Figure 4.6 presents a word cloud generated from the review text. Commonly occurring words such as *recipe*, *good*, *make*, and *food* are prominently displayed, reflecting the primary themes discussed by users. The word cloud provides a high-level overview of frequently used vocabulary, but is used only for exploratory purposes and not for model training. It helps in understanding general discussion patterns before applying sentiment classification models.

### 4.3 Machine Learning Model Performance

This section evaluates the performance of the supervised machine learning models used for sentiment classification. Logistic Regression and a Linear Support Vector Machine (LinearSVC) classifier were trained using contextual embeddings generated by a pretrained BERT model. Model performance was evaluated on unseen test data using accuracy, precision, recall, F1-score, and confusion matrices.

## Logistic Regression Performance

The Logistic Regression model achieved an overall accuracy of 96%, showing strong performance in classifying recipe reviews into negative, neutral, and positive sentiment categories. The model performed particularly well for positive sentiment, with a precision of 1.00 and a recall of 0.96,

indicating that positive reviews were identified very reliably. This can be partly explained by the clear language often used in positive reviews and the higher number of positive samples in the dataset. For neutral sentiment, the model achieved a high recall of 0.93, meaning that most neutral reviews were correctly identified. However, the precision for this class was lower (0.67), suggesting that some reviews predicted as neutral actually belonged to other sentiment categories. This highlights the challenge of identifying neutral sentiment, which often uses wording similar to that of mildly positive or negative opinions. The model also performed well on negative sentiment, with a recall of 0.89 and a precision of 0.84. Despite fewer negative examples, the model effectively captured negative sentiment patterns. In addition to the weighted F1-score of 0.96, the macro-averaged F1-score was used to ensure balanced evaluation across all sentiment classes, particularly given the imbalanced dataset. Overall, these results indicate that Logistic Regression performs consistently and reliably when used with BERT embeddings.

### **Linear Support Vector Machine (Linear SVC) Performance**

The Linear SVC model achieved a slightly higher overall accuracy of 97%, representing a modest improvement over Logistic Regression. This model is well-suited to high-dimensional feature spaces such as BERT-generated embeddings, as it focuses on learning clear decision boundaries between sentiment classes. For negative sentiment, Linear SVC achieved a precision of 0.87 and a recall of 0.85, indicating fewer false-positive predictions compared to Logistic Regression. This suggests that the model was more cautious when classifying reviews as negative. The neutral sentiment class also showed improved balance, with a precision of 0.74 and a recall of 0.90, reflecting better handling of neutral reviews by reducing confusion with other sentiment categories. The positive sentiment class continued to perform strongly, achieving a precision of 0.99 and a recall of 0.98, which confirms that Linear SVC effectively captures strong positive sentiment patterns. In addition to a weighted F1-score of 0.97, the macro-averaged F1-score was considered to ensure balanced performance across all sentiment classes. Overall, these results indicate that Linear SVC provides slightly better generalisation and more balanced classification performance when used with BERT embeddings.

```

... Logistic Regression Confusion Matrix:
[[ 190  20   3]
 [   8 222  10]
 [  28  88 3068]]
Logistic Regression Classification Report:
              precision    recall  f1-score   support

   negative      0.84      0.89      0.87      213
   neutral      0.67      0.93      0.78      240
   positive      1.00      0.96      0.98     3184

   accuracy      0.96
  macro avg      0.84      0.93      0.87      3637
 weighted avg      0.97      0.96      0.96      3637

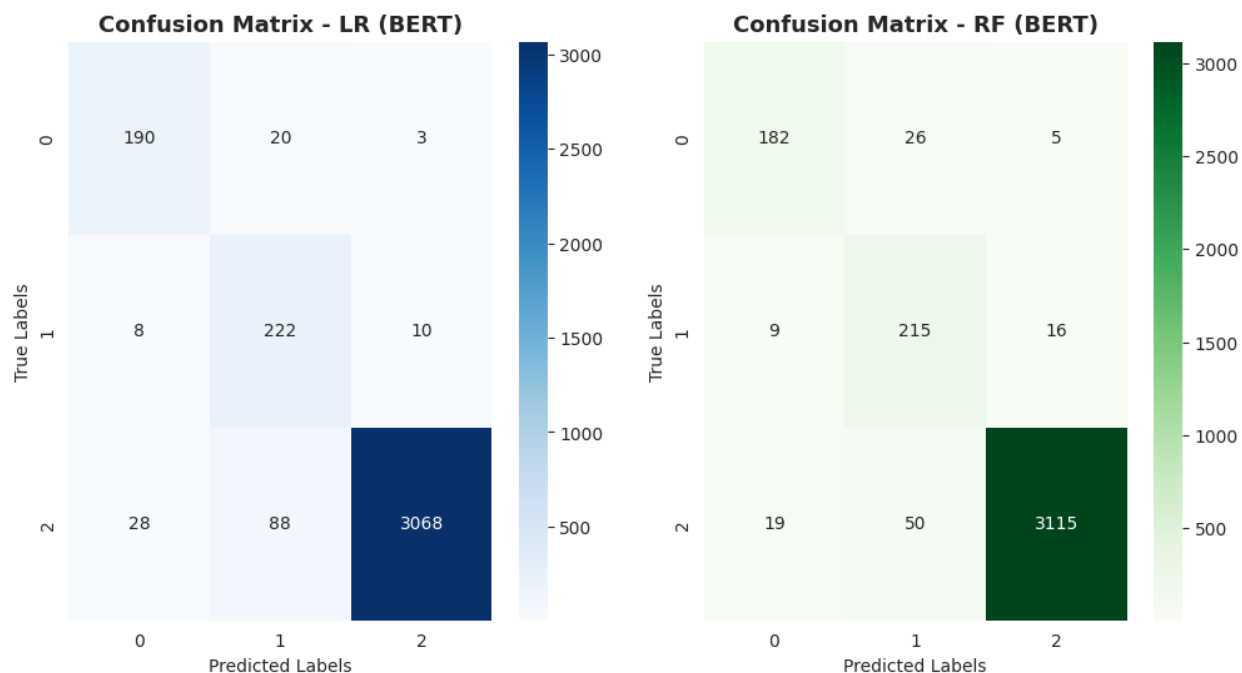
Linear SVC Confusion Matrix:
[[ 182  26   5]
 [   9 215  16]
 [  19  50 3115]]
Linear SVC Classification Report:
              precision    recall  f1-score   support

   negative      0.87      0.85      0.86      213
   neutral      0.74      0.90      0.81      240
   positive      0.99      0.98      0.99     3184

   accuracy      0.97
  macro avg      0.87      0.91      0.89      3637
 weighted avg      0.97      0.97      0.97      3637

```

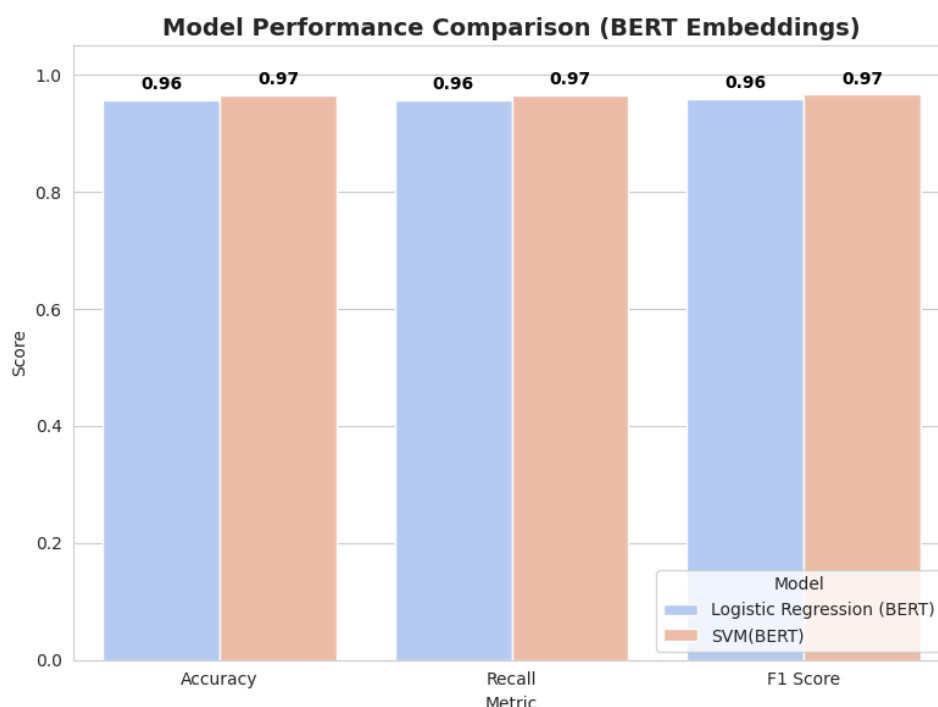
**Fig 4.7 Classification Report for Logistic Regression and Linear SVC**



**Fig 4.8 Confusion Matrix for Logistic Regression & Linear SVC**



## Model Comparison



**Figure 4.9: Model Performance Comparison Chart**

(Source: Created in Google Colab)

Both Logistic Regression and Linear Support Vector Machine models were evaluated using BERT-based embeddings. As Linear SVC achieved slightly higher classification accuracy and more balanced performance across sentiment classes, it was selected as the final model and saved for deployment in the sentiment analysis system.

To assess model generalisation and potential overfitting, training and testing accuracies were compared for both models. The slight difference between training and test performance indicates minimal overfitting and good generalisation to unseen data. This suggests that the models learned meaningful sentiment patterns rather than simply memorising the training dataset.

```

Train Accuracy: 0.9678239944998281
Test Accuracy: 0.956832554302997
Overfit Gap: 0.010991440196831181

```

**Fig 4.10 Overfitting Gap for Logistic Regression**

```

Train Accuracy: 0.9678239944998281
Test Accuracy: 0.956832554302997
Overfit Gap: 0.010991440196831181

```

**Fig 4.11 Overfitting Gap for Linear SVC**

## 4.4 Discussion and Implications

This study shows that combining BERT-based embeddings with classical machine learning models is a practical approach for sentiment analysis of recipe reviews. Both Logistic Regression and Linear SVC performed well, demonstrating the benefit of contextual embeddings over traditional text features. The results also highlight a strong positivity bias in user reviews, which is common on online platforms. Despite this imbalance, the models were able to identify negative and neutral sentiment with reasonable accuracy. The simulated real-time dashboard further illustrates how sentiment predictions can be visualised and interpreted effectively, making it a valuable approach for practical applications such as recipe platforms and customer feedback analysis. Overall, the study highlights the practical value of integrating Natural Language Processing, machine learning, and visualisation techniques to efficiently and interpretably analyse customer sentiment.

## 4.5 Summary

This chapter presented the findings and analysis of the developed sentiment analysis system. Exploratory data analysis highlighted a strong dominance of positive sentiment, along with clear trends in ratings and user engagement. Both Logistic Regression and Linear Support Vector Machine performed well when trained on BERT-based embeddings. Linear SVC achieved slightly higher performance and was selected as the final model, while Logistic Regression served as a

strong baseline. The simulated real-time dashboard effectively transformed model predictions into clear visual insights, demonstrating the practical feasibility of sentiment monitoring. These findings support the conclusions and future work discussed in the next chapter.

## **Chapter 5: Conclusion and Future Work**

This chapter summarises the key findings of the study, discusses their relevance to the research objectives, and outlines the implications, limitations, and directions for future work.

### **5.1 Summary of Key Findings**

Exploratory analysis revealed that the recipe review dataset is highly imbalanced, with positive sentiment dominating both review text and star ratings. This reflects generally high user satisfaction, a typical pattern on online review platforms. Logistic Regression and Linear Support Vector Machine models trained on BERT-based embeddings achieved strong performance, demonstrating the effectiveness of contextual embeddings for sentiment classification. The linear SVC achieved slightly higher accuracy and was selected as the final model.

The simulated real-time dashboard successfully transformed sentiment predictions into clear visual insights, enabling users to enter new reviews and immediately see sentiment outcomes. Overall, the results confirm that integrating NLP, machine learning, and visualisation provides a practical approach for analysing customer sentiment.

### **5.2 Alignment of Findings with Research Objectives**

#### **Objective 1: To preprocess and analyse textual feedback using NLP and sentiment classification models**

This objective was achieved by applying text preprocessing techniques, including cleaning, tokenisation, stopwords removal, and lemmatisation, followed by contextual embedding generation with a pretrained BERT model. Logistic Regression and Linear SVC classifiers were then used to predict sentiment, achieving high classification accuracy and confirming the effectiveness of the NLP pipeline.

#### **Objective 2: To develop a simulated real-time dashboard to visualise sentiment trends and customer insights**

The dashboard successfully displayed sentiment distributions, recipe-level comparisons, temporal trends, and word clouds. Although real-time data streaming was simulated, the dashboard

demonstrated how sentiment insights can be presented in an intuitive, interactive manner, effectively fulfilling the objective of visualising customer sentiment.

### **Objective 3: To evaluate model performance and system reliability**

Model evaluation using accuracy, precision, recall, F1-score, confusion matrices, and overfitting checks confirmed that both models generalised well to unseen data. The small gap between training and test accuracy indicated minimal overfitting, demonstrating the sentiment classification system's reliability and robustness.

### **Objective 4: To provide practical insights for decision-making**

The analysis identified recurring sentiment patterns, recipe popularity trends, and variations in customer feedback over time. These insights can support data-driven decision-making for content improvement, customer engagement strategies, and platform optimisation.

## **5.3 Practical Implications**

The outcomes of this study have several practical implications:

- Sentiment analysis can help identify recipes that are consistently well-received and those that require improvement.
- Platform administrators can monitor sentiment trends to detect emerging dissatisfaction and respond proactively.
- Positive sentiment insights can support content promotion and recommendation strategies.
- The dashboard framework demonstrates how sentiment intelligence can be integrated into consumer-facing applications to enhance user experience.

Overall, the system illustrates how sentiment analysis can support operational and strategic decisions in review-based platforms.

## **5.4 Limitations**

Despite the overall strong performance of the sentiment analysis system, several limitations were identified in this study. The dataset exhibited a significant class imbalance, with a strong dominance of positive reviews. Due to this positive skew, the models showed reduced precision when predicting neutral sentiment, as neutral reviews often share linguistic characteristics with

both positive and negative classes. This made neutral sentiment more challenging to distinguish accurately, even when using contextual BERT embeddings. Another limitation is that the system's real-time functionality was simulated using manual user input rather than live data streams. Reviews were entered manually into the dashboard, which limits the system's ability to reflect continuous, real-world data flow. Although this approach was sufficient to demonstrate real-time sentiment prediction behaviour, it does not fully represent large-scale deployment scenarios. In addition, the study focused on linear machine learning classifiers and did not explore fine-tuned transformer-based deep learning models due to computational constraints. The analysis also relied solely on textual data.

## **5.5 Future Work**

Future research could explore fine-tuning transformer-based models such as BERT or RoBERTa further to improve sentiment classification, particularly for minority classes. Integrating live data APIs would allow the dashboard to operate in a fully real-time environment. Additional data sources, such as images or cooking videos, could be incorporated to support multimodal sentiment analysis. Finally, sentiment trend forecasting could be implemented to enable proactive monitoring and decision-making.

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